Final Technical Summary

P09 Visual Similarity Search on Oxford IIIT Pet

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1 Problem and dataset

Problem: Build image search that returns visually similar pets, then compare self-supervised and supervised embeddings in terms of **retrieval quality and inference cost**.

Dataset: Oxford-IIIT Pet. We divide official trainval equally within breeds into gallery and validation queries, retaining official test queries for final evaluation. A committed manifest fixes disjoint image IDs. **Relevance:** query and neighbor must share a breed label.

Images	Breeds	Gallery	Validation	Test	Split seed
7,349	37	1,840	1,840	3,669	42

2 Methodology

Models: DINO and DINOv2 provide self-supervised representations; DeiT provides supervised baselines at small and base scales. Backbones are **pretrained and frozen**, with classifier heads excluded and no fine-tuning.

Pipeline: apply the same RGB conversion, bicubic resize, center crop and ImageNet normalization to every image. L2-normalize CLS features and rank the gallery by **exact cosine similarity**. Breed labels are used only for evaluation.

3 Experimental setup

Setting	Configuration
Image transform	Resize shorter side to 256 px; center crop 224 by 224 px; no augmentation.
Feature dimensions	DINO / DeiT-S: 384; DINOv2 / DeiT-B: 768.
Execution	Kaggle, Tesla T4, FP32; gallery batch 64; loader workers 2.
Timing sample	3 repeats; same 200 test queries; query batch 1; warm-up queries 10.

Metrics: Precision measures the relevant fraction of the returned set; **Recall** measures coverage of all relevant gallery images; **Hit rate** records whether any relevant neighbor is found. All scores are averaged over queries.

4 Main quality results

Official test results in percent. Higher is better; the best model is highlighted.

Model	P@5	P@10	R@5	R@10	Hit@5
DINO ViT-S/16	84.16	81.56	8.47	16.41	97.47
DeiT-S/16 supervised	86.65	85.27	8.72	17.16	96.92
DINOv2 ViT-B/14	**90.64**	**88.81**	**9.12**	**17.87**	**98.20**
DeiT-B/16 supervised	87.52	86.04	8.80	17.31	96.78

Finding: DINOv2 leads on every test metric; the validation ranking agrees. However, supervised DeiT-S outperforms the original DINO: **self-supervision does not guarantee better retrieval**.

5 Runtime results

Median across repeated benchmarks, after warm-up and CUDA synchronization. Query components use the median of per-repeat medians.

Model	Gallery build s	Preprocess ms	Embedding ms	Search ms
DINO ViT-S/16	8.14	5.60	6.25	0.224
DeiT-S/16 supervised	7.83	5.58	5.10	0.226
DINOv2 ViT-B/14	28.16	5.64	18.64	0.252
DeiT-B/16 supervised	20.62	5.66	12.47	0.253

Conclusion: DINOv2 offers the strongest quality, while DeiT-S is the faster embedding option. Gallery build includes image loading and transforms; query preprocessing is separate. Component medians are **not measured end-to-end latency**.

6 Failure analysis and limitations

Failures: DINO confuses an Abyssinian with other cat breeds; DINOv2 retrieves Birman images for a Siamese; DeiT-B retrieves Wheaten Terriers for a Leonberger. Pose and coat resemblance are plausible explanations, not proven causes.

Limits: breed agreement only approximates visual similarity. Architecture and pretraining data confound supervision comparisons. We did not test large galleries, audit near-duplicates or estimate confidence intervals; test reuse and illustrative case selection limit generalization. Failure figures are reused from an earlier run; their saved neighbors are checked.

7 Working system and reproducibility

Try the demo: upload your own photograph or choose a prepared example. Switch encoders or compare their results, then inspect retrieved pets, breed labels, similarity scores and live timings. See demo/ for setup.

Repeat the experiments: run the evaluation notebooks to extract embeddings and recalculate quality and runtime. Use the final analysis notebook to rebuild comparison tables and charts and review saved failures. See notebooks/.

Instructions: Detailed setup, required assets, environment and execution steps are documented in the repository: github.com/kite121/Visual_retrieval_system

8 Team contributions

Member	Contribution
Azamat	Dataset pipeline and DINO / DeiT-S baselines.
Karim	DINOv2 / DeiT-B integration and model comparison.
Aslan	Consolidated results, experiment verification and failure analysis.
Danil	Protocol, reproducible notebooks, repository integration, demo and final report.